

Question	Answer	Marks
5(a)	distance moved by wavefront/energy during one cycle/oscillation/period (of source) or <u>minimum</u> distance between two wavefronts or distance between two <u>adjacent</u> wavefronts	B1
5(b)	$(T =) 2.0 \times 2.5 (= 5.0 \text{ ms})$ or $2.0 \times 2.5 \times 10^{-3} (= 5.0 \times 10^{-3} \text{ s})$	C1
	$f = 1 / (5.0 \times 10^{-3})$ $= 200 \text{ Hz}$	A1
5(c)(i)	(path difference =) $8.0 + (20.8^2 - 8.0^2)^{0.5} - 20.8 = 6.4 \text{ (m)}$	A1
5(c)(ii)	• <u>path difference</u> = 4λ • waves (meet at C) in phase • constructive interference (of waves) <i>any two points, one mark each</i>	B2
5(c)(iii)	$v = 200 \times 1.6$ $= 320 \text{ (ms}^{-1}\text{)}$	C1
	$\Delta t = 6.4 / 320$ or $27.2 / 320 - 20.8 / 320$ $= 0.020 \text{ s}$	A1
5(c)(iv)	$3\lambda = 6.4$ $\lambda = 2.1 \text{ m}$	A1